
A Student's Proof Pamphlet

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Preface

The first noticeable fact about mathematics is that it fundamentally deals with abstract ideas rather than with tangible objects. For a preliminary illustration of this point, it suffices to consider a subject with which we are all familiar: arithmetic. There we perform operations on numbers without worrying about how to relate them in each case to concrete objects. The number five may of course appear in very concrete settings: there may be five sheep in a field, five books on a shelf, or five players on a team. But five itself is not a tangible object. We cannot see or touch it any more than we can see or touch time. It is an abstraction, a property shared by all collections containing as many elements as there are fingers on a hand. And of all such collections it is true that, if we add one more element, we get a collection of six. And if we add one more, we obtain a collection of seven. From this simple process of adding ones we can construct arbitrarily large numbers: a number larger than the number of people on Earth, larger than the number of grains of sand on Mars, larger than the number of atoms in the universe. That in practice counting is limited is not relevant. An abstraction is made from it, and from this abstraction the notion of infinity – an idea with no physical counterpart – enters quite naturally.

Abstractions of this sort are characteristic of the whole of mathematics, which is to say that mathematical objects are *imaginary*. They exist, so to speak, only in the playground of our imagination. There is no material from which mathematicians can carve their theorems; their imagination must supply both the stone and the chisel. It is true that mathematicians also make constant use of physical analogues to assist them in the discovery of their theorems, but even if we could measure the sides of a thousand different right-angled triangles with extreme accuracy, and found in each case that the square of the hypotenuse equaled the sum of the squares of the other two sides, we would still not have proved the Pythagorean theorem. To prove a theorem means for the mathematician to deduce it by logical argument from the fundamental properties of the concepts involved. Logic is therefore the final court of appeal in mathematics.

Unfortunately, the rules of the “legal procedure” are unknown to many students as they pass from school to university. In school, mathematics is often presented as a collection of formulas, and the unspoken “didactical contract” between the teacher and the student is that those who memorize the formulas and apply them correctly will be rewarded. At university, that contract is rewritten. The answer, until now the whole of the task, becomes only a part of it. To know that a formula is correct is no longer sufficient; one must know why it is correct, and show it by proof. Understood rightly, this is a gain. Behind the new demand lies a mathematics richer than the one left behind, a new world to explore. But many students feel like foreigners and get lost.

The question, then, is what maps the students have to guide them. The books that offer one are of two kinds. The first sets aside a few pages, at the opening of a course in calculus or in linear algebra, for the methods of proof. This is better than silence, but it is slight. The methods of proof are themselves abstractions, and an abstraction is not mastered in an afternoon; it asks for the same patient attention we give to any other part of mathematics, and a few pages cannot supply it. The second devotes a whole coursebook to the matter. This is generous, for the present purpose, perhaps too generous. Such a coursebook tends to introduce all at once sets and relations and functions, quantifiers and the formal machinery of logic: a considerable apparatus, much of which the beginner does not yet need in order to follow a simple proof, still less to find one.

Between the few pages and the whole coursebook there is a space that neither quite fills. It is the space this pamphlet is meant to occupy. It treats the methods of proof at greater length than an introductory chapter permits and with fewer prerequisites than a dedicated coursebook demands. Its examples are drawn from mathematics with which readers are already familiar, so that their attention may rest on the reasoning and not on the objects reasoned about. Much of what I prove in these pages is elementary, perhaps crossing the border into the obvious. Still, my hope is that it may spur some curiosity, give a little taste for mental work, and some means of doing it.

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CHAPTER 1

Logic

1.1. Deduction

We draw conclusions constantly, and for the most part without noticing that we do so. An old story shows the process at work, and shows too how quickly it can run.

A bishop was once the guest of honor at a country party given by a duke. At one point the duke left his guests to order something from his servants, and the bishop, eager to entertain the remaining company, told them that long ago, when he was a newly ordained priest, the first person whose confession he heard was a multiple murderer of the vilest kind. The duke thereupon returned, clapped the bishop on the shoulder and said, “The bishop and I are very old acquaintances. In fact, I was the first person whose confession he heard.” The rest of the guests hastily left.

No one present had seen the duke commit any crime. No accusation had been made; the bishop had named nobody and the duke had confessed to nothing but an old acquaintance. Yet the guests were in no doubt, and their haste to be gone shows it. The two things they were granted – that the first confession the bishop heard was a murderer’s and that the duke was his first penitent – settled the matter by themselves. The conclusion was already contained in them, and the guests had only to put them together.

The same putting-together happens in humbler cases, though without the scandal that fixes this one to memory. All men are mortal; Socrates is a man; therefore Socrates is mortal. The first statement gathers all men into the larger company of mortals. The second places Socrates among men. But if every man is already inside the company of mortals, then Socrates, being a man, is inside it too; and this is only to say that he is mortal.

This is **deduction**: the passing from statements granted or assumed to another which they compel. The first are called the **premises**, the last the **conclusion**. In our examples, the conclusion was reached without recourse to the world – we needed neither to witness the duke’s crimes nor to wait for Socrates’s death. All we had to do was to make plain to

ourselves what we had already accepted. We needed only to accept the premises, and the conclusion was inescapable.

We did not bother ourselves with the question of whether the premises were actually true, which is a separate question and often no concern of logic at all. There is a distinction between *correct information* and *correct reasoning*. Ideally we want both our reasoning and our information to be correct, but a deductive argument tells us only what we are bound to accept once the premises are granted. A single such step, from premises to conclusion, is an **inference**. A chain of them, leading from premises we accept to a conclusion we had not foreseen, is a **proof**.

1.2. Statements

The premises of an argument, and its conclusion, are **statements**. A statement is a sentence that is either true or false – the one or the other, and not both. "The duke is a murderer" is a statement: though we may not know which of the two it is, one of the two it must be. "Seven is larger than five" is a statement, and so is "three is larger than nine" – the former true, the latter false.

Not every sentence is of this kind. "Turn off the light!" is a command; it is a request for action, not an assertion. One cannot call "Turn off the light!" true or false. "Is the window open?" is a question, and a question asserts nothing – it asks. Only sentences that assert something, that lay down how matters stand, are true or false, and only these will concern us.

From statements we can make further statements. Given "the light is lit" and "the window is open," we may say, "the light is lit *and* the window is open," or "the light is lit *or* the window is open," or, of the first alone, "the light is *not* lit." The small words that join or alter statements in this way – **not**, **and**, **or** – are worth pausing over, for the whole of what follows is built with them.

Begin with the simplest, which alters a single statement rather than joining two. To deny a statement is to assert that it is not so. If "the light is lit" is true, then "the light is not lit" is false; and if the first is false, the second is true. Denial reverses truth and falsehood, and reverses it in every case. In the standard way of notating this logic, P denotes statements and the symbol $\neg$ denotes "not." Each connective can be set out in a truth table, thus (where T stands for "true" and F stands for "false")

P	$\neg P$
T	F
F	T

Under P are listed the possibilities of truth and falsity combinations. Here we have only one statement, so there are only two possibilities. The second column shows the result for $\neg P$. Since putting “not” in front of a statement reverses its truth value, doing it twice restores the original statement: $\neg(\neg P)$ is again P . If P is “the light is lit,” then $\neg P$ is “the light is not lit,” which is to say “the light is unlit,” and, thus, $\neg(\neg P)$ is “the light is not unlit,” which is to say “the light is lit.”

The word “and” joins two statements into one and the joined statement is true only when both parts are true. One false part is enough to condemn the whole. Letting P and Q denote statements and $\wedge$ denote “and”, matters are thus

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Now, we have two statements, so there are four possible combinations, and the third column is the result for $P \wedge Q$. A T occurs only in the row where P and Q are both true; in all other cases, where at least one of P and Q is false, $P \wedge Q$ is false. This gives a picture of the meaning of $P \wedge Q$, which aligns with what everyone means by “and.”

The word “or” is less straightforward, for in ordinary speech it says two different things. When a meal comes with coffee or tea, it comes with one or the other, not both: the “or” shuts out the case where you take both. But when a post is declared open to any citizen or resident, a person is not turned away for being both; the citizen who also resides there is welcome. Here the “or” admits the case where both hold. The same small word thus sometimes excludes the both-case and sometimes includes it. Logic cannot proceed on the vaguenesses and ambiguities of ordinary speech, so it has to settle on one. Let $\vee$ denote “or”. $P \vee Q$ is counted true when P is true, when Q is true, and when both are true; it is false only when both fail. This is the inclusive “or”, the “or” of the citizen and resident rather than that of the coffee and tea. The choice may look arbitrary, but the reason for it is plain enough: a word used in proofs must have its meaning fixed in advance, and this is the meaning fixed upon. The truth table is thus

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Notice what the three words have in common. In each case the truth or falsehood of the compound statement is settled entirely by the truth or falsehood of its parts. To judge whether "the light is lit and the window is open" is true, I need know nothing of lights or windows beyond whether each part is true; the word "and" does the rest. The parts carry their truth or falsehood, and the connecting word turns these into the truth or falsehood of the whole.

There remains one connective, and it is the one we have been moving toward. Almost every theorem in mathematics has the form *if this, then that*: if a number is divisible by four, then it is even; if two sides of a triangle are equal, then so are two of its angles. The words that join the one to the other are *if...then*, and one might expect them to be settled as readily as "and" or "or". They are not. Though they look as plain as the rest, they do not behave as the rest do, and they are best given a section to themselves.

1.3. Conditional Statements

The connective we set aside is the **conditional**. A conditional statement joins two statements by the words *if... then*: "if it rains, the ground is wet." The statement following "if" is the **antecedent**; the statement following "then" is the **consequent**. Here the antecedent is "it rains" and the consequent is "the ground is wet." Continuing using P and Q to denote statements, we write $P \Rightarrow Q$, and read this as "if P , then Q " or " P **implies** Q ."

But what does it mean for $P \Rightarrow Q$ to be true? Looking at the conditional "if it rains, then the ground is wet," we readily accept that it is true when P and Q are both true, and false when P is true and Q is false. But staring at it, we are not told what to say about the cases when it does not rain at all. To approach the matter, let us consider a conditional statement phrased as a promise: a professor says to you "if you earn an A on the final exam, then I will give you an A for your final grade." Now we ask, in what event can we say that the professor lied? For to break the promise is to make it false, and to keep it is to leave it true; so the case in which the promise is broken is the case in which this conditional is false.

There are four possibilities. In the first, you earn an A on the final exam and you later receive an A for your final grade. The professor did as promised, meaning you were told the truth, so when P and Q are both true, so is $P \Rightarrow Q$. In the second, you earn an A on the final exam, but not as a final grade. This is the situation when P is true and

Q is false. In this case, the professor did not do as promised, so $P \Rightarrow Q$ is false. These are the two scenarios we were ready to accept, thus

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F

Next, assume that you did not earn an A on the final exam but even so got an A as your final grade. This is the case when P is false and Q is true. Did the professor break the promise made to you? The professor did not, for no promises were made about any grade other than A on the final exam. Therefore, $P \Rightarrow Q$ is not false, so it is true:

P	Q	$P \Rightarrow Q$
F	T	T

Finally, suppose you did not earn an A , neither at the final exam nor as a final grade, so P and Q are both false. Again the professor did not break any promise. The only promise that was made was in the event you got an A on the final exam. So $P \Rightarrow Q$ is true:

P	Q	$P \Rightarrow Q$
F	F	T

The promise fails, then, in only one case: when you get an A on the final exam but not as a final grade. This is the rule we are seeking. A conditional is false when its antecedent is true and its consequent false, and true in every other case. The complete truth table is thus

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Three of the four rows, then, come out true, and only one false. The conditional fails in the single case where the antecedent is true and the consequent false – where you earned the A and were denied it – and holds in every other. In particular it holds in both cases where the antecedent is false: if you did not earn the A on the final, then whatever grade you were given, the promise is not broken, and so the conditional stands. A promise says nothing about the case its condition never arises in, and cannot be broken there.

A mathematical statement of the same form makes the same kind of promise. "If a number is divisible by six, then it is divisible by three" undertakes only this: that no number divisible by six shall fail to be divisible by three. A number that is not divisible by six – four, say, or nine – is like a day it did not rain; the statement promised nothing

of it, and is not broken by it, whatever such a number turns out to be divisible by.

The implication $Q \Rightarrow P$ is called the **converse** of $P \Rightarrow Q$. The two statements are not the same. "If it rains, then the ground is wet" is one statement; "if the ground is wet, then it rains" is another, and the two do not stand or fall together. The ground may be wet because a hose has passed over it, in which case the first holds and the second fails. Sometimes, though, it happens that $P \Rightarrow Q$ and $Q \Rightarrow P$ are both true. When this happens we say that P and Q are **equivalent**, and we write $P \Leftrightarrow Q$. This is also expressed by the words *if and only if*.

What this means is that P and Q are true in exactly the same cases: wherever the one holds, so does the other, and wherever the one fails, so does the other. They need not be the same statement, nor say the same thing. A number is even if and only if its last digit is even – a fact about the whole number on the one side, a fact about a single digit on the other, and yet the two stand or fall together. Tell me a number is even, and I know its last digit is even; tell me its last digit is even, and I know the number is even. To be told either is to be told both. The truth table is therefore

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

1.4. Quantifiers

The symbols $\wedge$, $\vee$, $\neg$, $\Rightarrow$, and $\Leftrightarrow$ are used to deconstruct many English sentences into a symbolic form. The aim of this symbolism is not to alienate, but to emphasize that the notions of "true" and "false" can only have a precise meaning when the symbols employed are themselves precise, and language, as we have seen, can be vague. One hitherto unresolved ambiguity concerns the use of **quantifiers**. These are words like "all," "some," "any," "every," and "nothing." Various as they are, they do the work of just two. The first is the **universal quantifier** "all" (or "every"), written $\forall$. The other is the **existential quantifier** "at least one" (or "some"), written $\exists$. An example of each is

U : All politicians are dishonest,

E : At least one politician is dishonest.

The universal quantifier claims something of every case, and so it is false the moment a single case fails. One exception is enough, and we call such an exception a **counterexample**. The existential quantifier, on the other hand, asks only for a single case; it is true if one case holds, and false only when every case fails. Their negations are therefore

$\neg U$: At least one politician is honest,

$\neg E$: All politicians are honest.

Notice how the two quantifiers trade places under denial. The careless denial of "all politicians are dishonest" is "all politicians are honest"; the correct denial is "some politician is honest" – a single honest politician overturns the claim, and nothing more is required. The denial of an *all* is a *some*. And the other way around: to deny that at least one politician is dishonest is to say that not one is, that all are honest. The denial of a *some* is an *all*.

One more turn completes the picture. A statement may carry two quantifiers, and then their order is not free. "Every lock has a key that opens it" and "there is a key that opens every lock" are built from the same words, yet they are not the same statement. The first gives each lock its own key, and different locks may want different keys. The second demands a single master key that opens all locks at once. The first may hold while the second fails – every lock openable, yet no one key for all of them. When a "some" stands inside an "every," exchanging their order can change a modest truth into an extravagant one, and the two must be kept apart.

1.5. The Conditional Revisited

Recall that $P \Rightarrow Q$ is false only when P is true and Q is false – that is, $P \Rightarrow Q$ is false only when $P \wedge (\neg Q)$ is true. This means that $P \Rightarrow Q$ is true exactly when $P \wedge (\neg Q)$ is false. This gives us the following result.

$$P \Rightarrow Q \text{ if and only if } \neg(P \wedge (\neg Q)).$$

Let us agree to call it **the implication law**.

Now, suppose we have two statements A and B , and that $A \Rightarrow B$ is true. We know that this means that $\neg(A \wedge (\neg B))$ is true. The "and" requires that both component statements are true, so the order of them does not matter. Therefore, $\neg((\neg B) \wedge A)$ is true. We will relabel $\neg B$ as R and $\neg A$ as S . Since $\neg S$ is equal to $\neg\neg A$, which is just A , we can write $\neg((\neg B) \wedge A)$ as $\neg(R \wedge (\neg S))$. The implication law, therefore,

tells us that $R \Rightarrow S$. Substituting back $R = \neg B$ and $S = \neg A$, we obtain $\neg B \Rightarrow \neg A$. We have, thus, shown that

$$A \Rightarrow B \text{ if and only if } \neg B \Rightarrow \neg A.$$

The statement $\neg B \Rightarrow \neg A$ is called the **contrapositive** of $A \Rightarrow B$, and the result tells us that to prove $A \Rightarrow B$ we can instead prove $\neg B \Rightarrow \neg A$. A very simple example is “If someone is in Oslo, then they are in Norway.” The contrapositive form is “If someone is not in Norway, then they are not in Oslo.” This feels almost obvious: once you know a person is outside Norway, Oslo is automatically ruled out. And that establishes the original claim as well.

The intuition is that the contrapositive approaches the same claim from the other end. Instead of asking, “If P is true, must Q be true?”, we ask, “If Q is not true, could P still be true?” If the answer is no, then P must imply Q .

1.6. Epilogue

All of mathematics takes the form of deduction. In proving a mathematical statement, or even in solving a simple equation, we take information that is known or assumed and apply deductive logic to produce new pieces of information, but we do not ask whether the information we start with is true. It is the form, and not the content, that constitutes the proof. Put differently, we are not in the least concerned with the truth or otherwise of our assumption, but merely with the truth of our deductions.

When a statement has a proof we will call it a **theorem** or a **proposition** or a **lemma** or a **corollary**. Unfortunately, there is no clear distinction between these terms, and their usage often depends on the taste of the writer, but as a general rule we say that a theorem is an important result that will be used over and over again; a proposition is a result that is considered less important than a theorem; a lemma is typically a smaller result whose main purpose is to help us prove a proposition or a theorem; and a corollary is a result that is an immediate consequence of the proposition or theorem that has just been proved. In any event, the important thing is that a statement that carries any one of these names has a proof. A rose by any other name would smell as sweet.

Every mathematical theorem can be formulated as an implication $P \Rightarrow Q$. To prove it is to rule out the one case that would make it false: that P is true and Q is false. The way to do this is to assume P and show, through a series of inferences, that this leads to Q . We may think of P as representing *what we know* and Q as representing *what we want*

to know, and the proof is the passage from the one to the other. To prove an equivalence $P \Leftrightarrow Q$ is then to prove two implications, $P \Rightarrow Q$ and $Q \Rightarrow P$, and so to make the passage both ways.

The question now is how we can build such passages. What steps are permitted? What shapes can a proof take? Section 1.5 has given us a taste. The next chapter will give us more.

CHAPTER 2

Proof Techniques

2.1. Preamble

We will see proofs involving integers, so it could be useful for us to recall some simple facts about them, if only to agree on some notation and terminology. The **integers** are the numbers

$$\dots, -2, -1, 0, 1, 2, \dots$$

They are **closed** under addition, subtraction, and multiplication, but not under division. This means that if m and n are integers, then so are $m + n$, $m - n$, and $m \cdot n$, but not necessarily m/n (e.g. 1 divided by 2).

It is common to divide the integers into two groups: **even** and **odd**. An integer is even if it is twice some integer, that is, if it can be written as $2k$ for some integer k . Thus, 6 is even since $6 = 2 \cdot 3$; and so is 0 since $0 = 2 \cdot 0$. This is also expressed by saying that an integer is even if it is **divisible** by 2. An integer that is not divisible by 2 is called odd. This means that an integer is odd if it can be written as $2k + 1$ for some integer k . For example, 7 is odd since $7 = 2 \cdot 3 + 1$.

More generally, we say that an integer m divides n if $n = mk$ for some integer k . When m divides n , we write $m|n$, and when m does not divide n we write $m \nmid n$. Divisibility therefore implies that n/m indeed is an integer. For example, 3 divides 12 since $12 = 3 \cdot 4$, and so $12/3 = 4$, but 5 does not divide 12 because this would leave a fractional remainder

$$\frac{12}{5} = \frac{10 + 2}{5} = \frac{10}{5} + \frac{2}{5} = 2 + \frac{2}{5} = 2.4.$$

Every integer is divisible by 1 and itself, since $n = 1 \cdot n$. Some positive integers can be written as a product of two smaller positive integers, and some cannot. Six can: $6 = 2 \cdot 3$. Seven cannot, for the only positive integers whose product is seven are 1 and 7 itself. An integer greater than 1 is called **prime** if it is only divisible by 1 and itself, and **composite** otherwise. Thus, 7 is prime and 6 is composite. Every integer greater than 1 can be expressed as a product of primes

and the first ten prime numbers are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29.$$

Finally, the integers are equipped with a natural ordering without an upper or a lower bound. The ordering is given by

$$\dots < -3 < -2 < -1 < 0 < 1 < 2 < 3 < \dots$$

Fix attention now to those integers greater than 0, and choose among them as you please – one, or a handful, or a great many – provided only that you choose at least one. Among those you have chosen there is always a smallest. Simple as it is, this last fact will carry real weight in what follows.

2.2. Direct Proof

Recall from Chapter 1 that all propositions in mathematics are of the form *if P, then Q* – a hypothesis *P* and a conclusion *Q*. To prove *if P, then Q* is to show that whenever *P* holds, *Q* holds too. So we grant ourselves *P* – we assume it – and set out to reach *Q*. We owe nothing about the case where *P* is false; the conditional promised nothing there, as Chapter 1 showed.

What this means in practice is that we proceed by a series of inferences. We do not pass from the hypothesis to the conclusion in a single stride. We pass instead from the hypothesis to some statement that follows from it, and from that to another, and so onward until the conclusion stands before us. Each step must be licensed – by a definition, or by a fact already agreed. A proof is a chain of such steps, its first link the hypothesis and its last the conclusion, and the proposition is proved once every step between has been supplied.

Where should the steps begin? Almost always with the definitions. The first statement we shall prove – that the sum of two even integers is even – is a conditional in disguise, for it says that *if* two integers are even, *then* their sum is even. But the word even, taken by itself, offers nothing to work with. Its definition does: it tells us that each of the two integers is of the form $2k$ for some integer k . Written in this form the hypothesis can be handled by ordinary arithmetic, and the arithmetic, with luck, delivers the form named in the conclusion.

We turn to the examples.

Proposition 2.2.1

The sum of two even integers is even.

Proof. Let x and y be even integers. By the definition of an even number, $x = 2m$ for some integer m , and $y = 2n$ for some integer n . Then,

$$x + y = 2m + 2n = 2(m + n).$$

Now, $m + n$ is an integer, since the sum of two integers is again an integer. Thus, $x + y$ is twice an integer, which is to say it is even. $\square$

Observe what the last step did. We factored the 2 back out, not to simplify the arithmetic – $2m + 2n$ was simple enough – but to make the form of an even number visible in the result. The proof begins by writing the definition in and ends by reading the definition out.

Observe, too, the letters. We did not verify that $2 + 4$ is even, and that $6 + 8$ is even, and grow more confident with each case. Such checks would no more prove the statement than measuring a thousand right-angled triangles would prove the theorem of Pythagoras. The letter m can be any integer we would like, and so can n ; in proving the result for m and n we have proved it for every pair of even integers at once. This is why the proofs that follow are written in letters where a calculation would use numbers. The little square at the end of the proof is called a **halmos** and means *quod erat demonstrandum* (sometimes abbreviated qed or QED) or “what was to be demonstrated,” and signals that the proof has been completed.

Proposition 2.2.2

If n is odd, then n^2 is odd.

Proof. Suppose n is odd. Then, $n = 2k + 1$ for some integer k , and so

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 has the form $2m + 1$ where $m = 2k^2 + 2k$, and is therefore odd. $\square$

The shape of this argument is the same as before, but the last step asks a little more of us. In the first proof, the form of the conclusion $2(m + n)$, lay ready on the page. Here we must arrange the result deliberately before the form of an odd number can be seen. Recognizing the conclusion is not always automatic; sometimes the result must first be coaxed into the required shape. We shall have occasion to use this statement again in a later section.

Proposition 2.2.3

Let a , b and c be integers. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof. Assume that $a \mid b$ and $b \mid c$. Writing each hypothesis in the form given by the definition of divisibility, we have $b = am$ and $c = bn$ for some integers m and n . Inserting the former into the latter,

$$c = bn = (am)n = a(mn).$$

Since mn is an integer, c is a times an integer, and so $a \mid c$. $\square$

This proof has two hypotheses rather than one, and its central step is to substitute the one into the other. It also carries the method away from even and odd numbers, where our first two proofs kept it, and applies it to divisibility instead. The technique does not depend on the subject. Whatever the definitions are, one writes the hypotheses in the form those definitions supply, works forward by ordinary steps, and recognizes the conclusion when it appears.

For our last example the hypothesis cannot be carried forward in one piece. It must first be separated into cases.

Proposition 2.2.4

Let n be a positive integer. Then, $1 + (-1)^n(2n - 1)$ is a multiple of 4.

Before proving this, we should see where the difficulty lies. The expression contains the factor $(-1)^n$, whose value is not fixed: it is 1 when n is even and -1 when n is odd. Until we know which, we cannot simplify the expression, for the two possibilities lead to different arithmetic. But every integer is either even or odd, and none is both. So we consider the two possibilities in turn, prove the claim under each, and, since no third possibility remains, conclude that it holds in every case.

Proof. We divide into two cases according to whether n is even or odd. Suppose first that n is even. Then, $(-1)^n = 1$, and

$$1 + (-1)^n(2n - 1) = 1 + (2n - 1) = 2n.$$

Since n is even, $n = 2k$ for some integer k , and so the expression equals $2(2k) = 4k$, which is a multiple of 4.

Suppose now that n is odd. Then, $(-1)^n = -1$, and

$$1 + (-1)^n(2n - 1) = 1 - (2n - 1) = 2 - 2n.$$

Since n is odd, $n = 2k + 1$ for some integer k , and so $2 - 2n = 2 - 2(2k + 1) = -4k = 4(-k)$, again a multiple of 4. $\square$

A word in closing. In each of these proofs we did the same four things. We assumed the hypothesis. We wrote it in the form its definition provides. We worked forward by ordinary arithmetic, each step licensed by what we had already granted. And we recognized, in what we reached, the form named by the conclusion. The definitions did the heavy work: they are what turned words we could only recognize into forms we could calculate with. When one is faced with a statement to prove and does not see the way in, the definitions are almost always the place to begin.

2.3. Proof by Contraposition

The last section closed on a piece of advice: when faced with a statement to prove, begin with the definitions. Write the hypothesis in the form its definition supplies, and work forward. For every proof in that section the advice served. We should not expect it always to.

Consider the statement “if n^2 is even, then n is even”. Let us try to prove it as we proved the others. We assume the hypothesis and write it in its defining form: n^2 is even, so $n^2 = 2k$ for some integer k . Now we are to reach the conclusion, that $n = 2m$ for some integer m . But from $n^2 = 2k$ there is nowhere to go. The equation tells us what n^2 is, and the conclusion asks about n ; and no rule of arithmetic lets us pass from the one to the other. We can deduce that $n = \sqrt{2k}$, but with this expression it is not clear that n is an integer, let alone an even one. The definition of the hypothesis, written out, offers us nothing to work with. The direct method stalls at its first step.

But what if the roles were exchanged, if we could assume something about n and be asked to conclude something about n^2 ? Then, the direct method would have something to bite on, for from n we can always form n^2 .

If you have not read section 1.5, this would be a good time, for it tells us exactly what to assume. To prove “if n^2 is even, then n is even” we can instead prove “if n is not even, then n^2 is not even.” Or symbolically, to prove $P \Rightarrow Q$ we can instead prove $\neg Q \Rightarrow \neg P$. This is **proof by contraposition**.

Proposition 2.3.1

Let n be an integer. If n^2 is even, then n is even.

Proof. We prove the contrapositive: if n is not even, then n^2 is not even – that is, if n is odd, then n^2 is odd. But this exactly what we proved in Proposition 2.2.2. The contrapositive therefore holds, and with it the statement itself. $\square$

This first example shows the method at its plainest. The swap alone did the work: the statement we could not begin with was equivalent to one we had already finished. Nothing remained but to notice it.

Proposition 2.3.2

If $3n + 2$ is odd, then n is odd.

Proof. We prove the contrapositive: if n is not odd, then $3n + 2$ is not odd – that is, if n is even, then $3n + 2$ is even. Suppose, then, that n is even. Then, $n = 2k$ for some integer k , and

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

Since $3k + 1$ is an integer, $3n + 2$ is even. This proves the contrapositive, and therefore the original statement. $\square$

Here the contrapositive was not a result already in hand, and we proved it outright. Notice that we did nothing a reader of Section 2.2 has not done many times: we wrote the hypothesis in its defining form, carried out the arithmetic, and recognized the conclusion. The work was a direct proof from beginning to end. Only its subject – the contrapositive rather than the original – was chosen with care.

For the last example the hypothesis and the conclusion are compound, and forming the contrapositive requires more attention.

Proposition 2.3.3

Let a and b be integers. If ab is even, then a is even or b is even.

Proof. We prove the contrapositive. The negation of ab is even is ab is odd. The negation of a is even or b is even is that none of them are even. The contrapositive therefore reads if a is odd and b is odd, then ab is odd. Suppose, then, that a and b are both odd. Then, $a = 2m + 1$

and $b = 2n + 1$ for some integers m and n , and

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n$ is an integer, ab is odd. This proves the contrapositive, and therefore the original statement. $\square$

The proof itself was routine; the care lay in stating what was to be proved. Had we carelessly taken the negation of "a is even or b is even" to be "a is odd or b is odd," we should have set out to prove the wrong thing. Forming the contrapositive is a logical operation, and the rules of Chapter 1 govern it.

Let us be clear about what we have and have not done. We have not introduced a new way of reasoning. Every proof in this section, once its opening line was written, proceeded exactly as the proofs of Section 2.2 did – by assuming a hypothesis and working forward to a conclusion. What is new is a choice made before the working begins: to prove, in place of the given conditional, the equivalent conditional got by negating both parts and reversing them. We make this choice when the hypothesis, written in its defining form, gives us nothing to calculate with, while the negated conclusion does. Nothing in such a proof is supposed false, and nothing is driven to an absurdity; the argument is direct throughout. This is worth fixing in mind, for the next section presents a method with which contraposition is easily confused, and the difference between them is exactly the point.

2.4. Proof by Contradiction

In a direct proof of $P \Rightarrow Q$, we assume P and work forward until we arrive at Q . In a contrapositive proof of $P \Rightarrow Q$, we assume $\neg Q$ and work our way forward until we arrive at $\neg P$. Now, recall once more that $P \Rightarrow Q$ fails in exactly one circumstance: when P is true and Q is false. Let us therefore assume both P and $\neg Q$, and suppose that this assumption leads us to the conclusion that some statement C and its negation $\neg C$ are both true. A statement cannot be true *and* false at the same time; $C \wedge (\neg C)$ is always false. So we have shown that $P \wedge (\neg Q) \Rightarrow C \wedge (\neg C)$ is true while its conclusion $C \wedge (\neg C)$ is false. But a true implication cannot have a false conclusion unless its hypothesis is false as well (see the final row of the truth table on page 5). Hence, $P \wedge (\neg Q)$ is false, so $\neg(P \wedge (\neg Q))$ is true, which by the implication law of Section 1.5 means that $P \Rightarrow Q$.

What we have argued here is this: to prove $P \Rightarrow Q$, we can assume P and $\neg Q$ and show that this leads to some contradiction – a statement

of the form $C \wedge (\neg C)$, such as “ $0 = 1$ ” or “ n is even and odd” or “ $x \neq y$, yet $(x - y)^2 \leq 0$ ”. This is **proof by contradiction**.

Before the examples, two remarks are worth making. The first concerns the warning that closed the last section: this method is easily confused with contraposition, and telling them apart is the whole point. So let us be crystal clear. Contraposition assumes $\neg Q$, and $\neg Q$ alone; its target is fixed: it must reach $\neg P$. Contradiction grants us both $\neg Q$ and P , and its target is a contradiction $C \wedge (\neg C)$, no matter which. That openness is the strength of the method, and, as the second remark observes, also its hazard.

For it is not always possible to say beforehand what the contradiction will be, and this costs us a safeguard that the direct method silently provides. In a direct proof, a false step usually leaves us stranded: the conclusion we are marching toward stays out of reach, and the failure warns us that something is wrong. A proof by contradiction, on the other hand, gives us no such warning. Its goal is merely to reach some statement that clashes with itself, and a false step is fully capable of producing one. A proof by contradiction therefore asks for more vigilance, not less.

We turn to the examples.

Proposition 2.4.1

No odd integer is the sum of three even integers.

Proof. Assume for the sake of a contradiction that there exists an odd integer n that can be expressed as the sum of three even integers a , b and c . Then, there are integers x , y and z such that $a = 2x$, $b = 2y$ and $c = 2z$. Hence,

$$n = a + b + c = 2x + 2y + 2z = 2(x + y + z).$$

But $x + y + z$ is an integer, so this means that n is even. Hence, n is both odd and even. We have reached a contradiction; the statement holds. $\square$

This is an example of the kind of statement where a proof by contradiction is useful: a statement asserting that something cannot happen – that no object of a certain description exists. We supposed the object existed, and the supposition undid itself. The square root of 2 furnishes a claim of the same shape, and a more famous one: not that a certain sum cannot occur, but that a certain number cannot be written as a fraction.

Proposition 2.4.2

$\sqrt{2}$ is irrational.

Proof. A number is rational if it can be written as a fraction p/q of two integers with $q \neq 0$. Assume for the sake of a contradiction that $\sqrt{2}$ is rational, so that

$$\sqrt{2} = \frac{p}{q}$$

for some integers p and q with $q \neq 0$. We may assume the fraction is in lowest terms – that p and q have no common factors – for if they had one, we could divide it out and obtain a fraction of the same value with smaller terms. Squaring both sides gives $2 = p^2/q^2$, and hence

$$p^2 = 2q^2.$$

This means that p^2 is even, and, by Proposition 2.3.1 so is p . Write $p = 2k$ where k is some integer. Then, $(2k)^2 = 2q^2$, that is, $4k^2 = 2q^2$, and hence,

$$q^2 = 2k^2.$$

This means that q^2 is even, and by Proposition 2.3.1 again, q is even. But now p and q are both even, and so both are divisible by 2, contradicting our assumption that they have no common factor. Therefore, $\sqrt{2}$ is irrational. $\square$

One of the most famous proofs in the history of mathematics is Euclid's proof of the infinitude of primes. Not including it here would be negligence.

Proposition 2.4.3

There are infinitely many primes.

Proof. Assume, for the sake of a contradiction, that there are only finitely many primes, say n in all, and list them in increasing order as $p_1, p_2, \dots, p_n$. Let

$$N = p_1 p_2 \cdots p_n$$

be their product, and consider $N + 1$. Being a positive integer greater than 1, $N + 1$ is either prime or composite.

If $N + 1$ is prime, then it is a prime absent from our list, for it is larger than p_n and so equals none of them, even though the list was supposed to contain every prime.

If $N + 1$ is composite, then some prime q divides it, so $N + 1 = aq$ for some integer a . We claim that q is not on our list. For suppose it were. Then, q must divide N as well, $N = bq$ for some integer b , and also their difference,

$$1 = (N + 1) - N = aq - bq = q(a - b),$$

which is impossible, since no prime divides 1. The supposition is untenable, and so q is not on our list after all. We have again found a prime the list omits.

Either way, the list fails to contain every prime, contradicting the assumption that it was complete. Therefore, there are infinitely many primes. $\square$

Each of these proofs supposed the very thing the theorem forbids – an odd integer built from three even ones, a fraction squaring to 2, a finite list of every prime – and followed the supposition until it collapsed. This is what contradiction does that the other methods often cannot: when a theorem denies that something exists, there is usually nothing to build toward, and the natural course is to assume the thing exists and watch it undo itself. One assumption is made, and one impossibility is reached, and between them the theorem is proved.

2.5. Mathematical Induction

Sometimes we want to prove that a statement $P(n)$ holds for every positive integer n . We cannot verify each one in turn, for they never end. But there is a structured method, **mathematical induction**, that consists of two steps:

- (1) Show that $P(1)$ is true.
- (2) Show that $P(k) \Rightarrow P(k + 1)$.

If these two conditions are met, the logic unfolds like this: the first step gives us $P(1)$. The second step, applied to $P(1)$, gives us $P(2)$; applied to $P(2)$, it gives us $P(3)$; applied to $P(3)$, $P(4)$; and so on. Every positive integer is reached in its turn. Induction works like an infinite line of dominoes: to know that all of them fall, we need only know that the first one falls, and that each falling domino knocks over the next.

The domino picture persuades, but we can prove that induction works, and the proof rests on a fact established in the last section. Suppose $P(1)$ and $P(k) \Rightarrow P(k + 1)$ are both true, and assume, for a contradiction, that $P(n)$ is false for some positive integer n . Then there is a positive integer for which P fails, and so, by the least-element

principle, a smallest one; call it m . Now, m cannot be 1, since $P(1)$ is true; so $m > 1$. Since $m - 1$ is an integer smaller than m , $P(m - 1)$ is true. But by assumption $P(m - 1) \Rightarrow P(m)$, and so $P(m)$ is true, contradicting that P fails at m . The assumption was therefore false, and $P(n)$ holds for every positive integer n .

Let us look at some examples.

Proposition 2.5.1

For all positive integers n ,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Proof. We argue by induction on n .

Base Case: If $n = 1$, the left side is just 1, and the right side is $1 \cdot 2 / 2 = 1$. The two agree, so the statement holds for $n = 1$.

Induction Hypothesis: Assume that the formula is true for some arbitrary positive integer k . That is, assume

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}.$$

Inductive Step: We must show that the formula holds for $k + 1$. That is, that

$$1 + 2 + 3 + \dots + k + (k + 1) = \frac{(k + 1)(k + 2)}{2}.$$

Begin with the left-hand side and set apart its final term

$$1 + 2 + 3 + \dots + k + (k + 1) = (1 + 2 + 3 + \dots + k) + (k + 1)$$

The sum in the first parentheses on the right side is exactly the left side of the inductive hypothesis, so we may replace it:

$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This is the right side for $k + 1$. So whenever the statement holds for k , it holds for $k + 1$. By the principle of mathematical induction, the formula holds for every positive integer n . $\square$

The whole of the inductive step turned on one move: separating the last term, so that the sum of the first k terms appeared, and could be replaced by what the hypothesis tells us it equals. This is the pattern of every induction. The step never proves the case $k + 1$ from nothing; it proves it from the case k , and the work is to make the case k appear.

Proposition 2.5.2: Bernoulli's Inequality

For all positive integers n , $(1 + x)^n \geq 1 + nx$ when $x > -1$.

Proof. Fix $x > -1$, and argue by induction on n .

Base Case: For $n = 1$, the left side is just $1 + x$, and the right side is $1 + x$. They are equal, so the inequality holds.

Induction Hypothesis: Assume that the inequality holds for some positive integer k :

$$(1 + x)^k \geq 1 + kx$$

Inductive Step: We must show that the formula holds for $k + 1$. That is, that

$$(1 + x)^{k+1} \geq 1 + (1 + k)x.$$

Write the left side as $(1 + x)^{k+1} = (1 + x)^k (1 + x)$. From the induction hypothesis, we have that $(1 + x)^k \geq 1 + kx$. Since $1 + x > 0$, we get

$$(1 + x)^{k+1} \geq (1 + kx)(1 + x).$$

Expanding the right side we get

$$(1 + kx)(1 + x) = 1 + (1 + k)x + kx^2.$$

Since $kx^2 \geq 0$, subtracting it from the right side will only make that side smaller, so

$$(1 + kx)(1 + x) \geq 1 + (1 + k)x.$$

Combining the two inequalities,

$$(1 + x)^{k+1} \geq 1 + (1 + k)x,$$

as desired. By the principle of mathematical induction, the formula holds for every positive integer n . $\square$

An inductive step for an inequality asks something the equation did not. When both sides are equal, we may rewrite freely and the equality is never in danger. An inequality must be *maintained* across the step, and it survives only if each operation respects its direction. So watch out for negative numbers.

Here is one more, written more compactly.

Proposition 2.5.3

For every positive integer n , the number $n^3 - n$ is divisible by 3.

Proof. We argue by induction on n . For $n = 1$, we have $n^3 - n = 1 - 1 = 0$, and 0 is divisible by 3, since $0 = 3 \cdot 0$. So the statement holds for $n = 1$. Next, assume that the statement holds for some positive integer k . This means that $k^3 - k = 3m$ for some integer m . We must show that 3 divides $(k + 1)^3 - (k + 1)$. Expanding this expression we get,

$$(k + 1)^3 - (k + 1) = k^3 + 3k^2 + 3k - k = (k^3 - k) + 3k^2 + 3k.$$

The induction hypothesis has appeared. Hence,

$$(k + 1)^3 - (k + 1) = 3m + 3k^2 + 3k = 3(m + k^2 + k).$$

Since $m + k^2 + k$ is an integer, $(k + 1)^3 - (k + 1)$ is three times an integer, and so it is divisible by 3. By induction, the statement holds for all positive integers n . $\square$

Once more: to use induction is to arrange the case $k + 1$, so that the case k shows up. The expression for $k + 1$ did not resemble the expression for k until we expanded it, but then, among the terms, it sat. Once it was isolated, the induction hypothesis could be applied to it, and what remained, $3k^2 + 3k$, was plainly a multiple of 3.

We have now seen four ways to prove that a mathematical statement is true. This is the basic apparatus of mathematics. Having the tools, though, is not the same as knowing which to reach for, and that is often the harder part. There is no rule for it. But there is a habit that helps. Faced with a statement to prove, one does well to ask whether one has faced a similar problem and solved it before. For what unlocks it is usually a bright idea, and bright ideas are the good fortune of past experiences. This is why proofs are learned by reading proofs. Each one studied is added to the store from which later ideas are drawn, and the more proofs one has seen, the more readily the next bright idea will come.

CHAPTER 3

Miscellaneous

3.1. A Short Introduction

If you have reached this point, you have, in principle, everything you need to get started. The four methods of the previous chapter are enough to prove a great many theorems, and nothing that follows replaces them. What follows is a small collection of things intended to make the next bright idea more likely to come.

Some of these are forced upon us by the form of what we wish to prove. To show that two statements are equivalent, we have no choice but to argue in both directions; the shape of the claim decides the shape of the proof before we begin. Others are not forced upon us at all. They are conveniences – small economies that a proof does not require but that a sensible person will take. We will see examples of both here, and we begin, as always, with something already familiar.

3.2. Existence and Uniqueness

The **real numbers** are the numbers we are accustomed to working with, numbers like 1, 2, $\sqrt{2}$, π , and their negatives. Children use them to count the number of fingers on their hand, carpenters use them to measure lengths and areas, and financial advisors use them in reckoning profits, losses, and percentages.

One of the first things one learns to do with real numbers is to solve linear equations. Consider

$$2x = 6.$$

The answer is $x = 3$. This is the number that when inserted for x on the left side gives the number on the right side.

Now, consider the equation

$$x^2 = 4.$$

This equation has solutions $x = 2$ or $x = -2$. Meanwhile, the equation

$$x^2 = -1,$$

has no solution among the real numbers. Thus, any equation really asks two questions. Is there a solution? And is that solution unique? As these are two separate questions, they are worth keeping separate.

Granting the real numbers the usual rules of arithmetic, we state the following result.

Proposition 3.2.1

The general linear equation $ax = b$ with $a \neq 0$, has a unique solution $x = b/a$.

Proof. We first show that $x = b/a$ is a solution:

$$a \cdot \frac{b}{a} = 1 \cdot b = b,$$

which settles existence.

We now show that this solution is unique. To this end, assume that x and y are both solutions. Then, $ax = b$ and $ay = b$. This means that $ax = ay$ and, hence, that $x = y$. Therefore, if x and y are both solutions, then $x = y$, so there is at most one solution. This completes the proof. $\square$

In other words, to show existence, we need only provide a particular example that shows it is true. To show uniqueness, we assume that two solutions exist and show they must be equal; that is, that there cannot be two different ones.

3.3. If and Only If

A statement of the form P if and only if Q asserts, as we have seen, two things: that $P \Rightarrow Q$ and that $Q \Rightarrow P$. To prove it we must, therefore, prove both things. However, they need not be proved in the same way. Sometimes the first half is best showed using one technique, and the second by using another.

Proposition 3.3.1

For any integer n , 3 divides n if and only if 3 divides n^2 .

Proof. For the forward direction, suppose 3 divides n . Then, $n = 3k$ for some integer k and, hence,

$$n^2 = (3k)^2 = 9k^2 = 3(3k^2),$$

so 3 divides n^2 .

For the reverse direction, we prove the contrapositive statement if 3 does not divide n , then 3 does not divide n^2 . Suppose that 3 does not divide n . Then, n must leave a remainder of 1 or 2 when divided by 3. That is, $n = 3k + 1$ or $n = 3k + 2$ for some integer k . If $n = 3k + 1$, then

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1,$$

so 3 does not divide n^2 . If $n = 3k + 2$, then

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1,$$

so again, 3 does not divide n^2 . This completes the proof. $\square$

3.4. Sets: Membership and Inclusion

A set is a collection of things. A pack of wolves, a bunch of grapes, or a team of players are all examples of sets. The things that make up the set are called its **elements** (or **members**). An element of a set can be a wolf, a grape, or a player, but we are interested in sets containing numbers. By convention, capital letters $A, B, C, \dots, X, Y, Z$ denote sets and lower-case letters $a, b, c, \dots, x, y, z$ denote elements of a set. If x belongs to a set A , we write

$$x \in A,$$

which reads “ x is an element of A ” or “ x is in A .” If x does not belong to A we write $x \notin A$.

Sometimes it is convenient to express sets by listing all elements that belong there between commas in braces. For example, $A = \{1, 2, 3, 7\}$ is a set containing the numbers 1, 2, 3, and 7. The most important way, however, to describe a set is by defining the property, or properties, that all members of the set must satisfy. A set that illustrates how this is done is

$$\{x : x \text{ is prime and } x < 20\},$$

which, in fact, is the set $\{2, 3, 5, 7, 11, 13, 17, 19\}$. Here the opening brace reads “the set of” and the colon reads “such that.” The set, thus, reads, “the set of all x such that x is prime and x is less than 20.” This way to describe a set is called **set-builder notation**.

From a given set B , assumed to contain at least one number, we can always form a new set A by picking some of the elements in B . The result is a set A where every element in A is also an element in B , so the set A is contained in the set B , or B includes A . This is denoted $A \subseteq B$.

The question that arises is how can we show that one set is a subset of another? By definition, $A \subseteq B$ means that any x belonging to A also

belongs to B . This tells us that to prove $A \subseteq B$, we pick an arbitrary $x \in A$ and show that this implies that $x \in B$. In practice, this usually means to grant x the properties characterizing elements of A , and use those properties to show that x also has the properties required to belong to B . This is called the **element argument method**.

Proposition 3.4.1

If

$$A = \{n : n \text{ is an integer and } 12|n\},$$

and

$$B = \{n : n \text{ is an integer and } 3|n\},$$

then $A \subseteq B$.

Proof. We must prove that if $x \in A$, then $x \in B$. So assume that $x \in A$. Then, $x = 12k$ for some integer k . But $12k = 3(4k)$ and since k is an integer, so is $4k$. Thus, $x = 3m$ where m is the integer $4k$. By definition, 3 divides x and, therefore, $x \in B$. Since $x \in A$ implies $x \in B$, it follows that $A \subseteq B$, and the proof is complete. $\square$

In summary, to prove that one set lies inside another is to prove a statement of the form “if x is in the first set, then x is in the second.” The subset relation is an implication in disguise. And should we ever wish to show that two sets are equal, we have only to show that each lies inside the other – two implications, one in each direction, exactly as with an *if and only if*.

3.5. Difference of Squares and the AM-GM Inequality

A square is never negative. Whatever real number x may be, $x^2 \geq 0$. This modest fact is the whole of a useful technique: to prove an inequality, carry everything to one side and try to recognize what remains as a square. For this example, we only need to know that $x^2 - y^2 = (x + y)(x - y)$.

Proposition 3.5.1

If x and y are positive numbers and $x \geq y$, then $\sqrt{x} \geq \sqrt{y}$.

Proof. Assume that x and y are positive numbers and that $x \geq y$. Since $x \geq y$, $x - y \geq 0$. And since $x = (\sqrt{x})^2$ and $y = (\sqrt{y})^2$, we can rewrite the

expression as

$$(\sqrt{x})^2 - (\sqrt{y})^2 \geq 0.$$

The left side is a difference of squares, so

$$(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq 0.$$

Since x and y are both positive, so is $\sqrt{x} + \sqrt{y}$, which allows us to divide both sides by $(\sqrt{x} + \sqrt{y})$. This gives

$$\sqrt{x} - \sqrt{y} \geq 0,$$

or simply $\sqrt{x} \geq \sqrt{y}$. □

This result can be used to prove the following inequality, which one on occasion will use again. It will be useful to know that $(x + y)^2 = x^2 + 2xy + y^2$.

Proposition 3.5.2: AM-GM Inequality

If x and y are positive numbers, then $\sqrt{xy} \leq \frac{x+y}{2}$.

Proof. Assume that x and y are positive numbers. Observe that $0 \leq (x - y)^2$ because the square of real number is always non-negative. Expanding the right side, we get

$$0 \leq x^2 - 2xy + y^2.$$

Adding $4xy$ to both sides gives

$$4xy \leq x^2 + 2xy + y^2,$$

or

$$4xy \leq (x + y)^2.$$

Since both sides are positive, we can apply Proposition 3.5.1, which, in turn, gives

$$2\sqrt{xy} \leq x + y.$$

Dividing by 2 proves the result. □

The proof actually shows a little more than the statement. If $x = y$, then the left side becomes $\sqrt{x^2} = x$ and the right side $2x/2 = x$, so it is an equality. Meanwhile, if $x \neq y$, then the one inequality the proof rests on, $0 \leq (x - y)^2$, is strict – meaning that $0 < (x - y)^2$ – and since that is the only inequality in the proof, its conclusion is strict too: $\sqrt{xy} < (x + y) / 2$.

A very natural application of the AM-GM inequality is the **rectangle problem**. Recall that a rectangle is a four-sided flat shape where all angles are right angles and opposite sides are parallel and of equal length. The perimeter is the distance around the edges.

Proposition 3.5.3

Among all rectangles with a fixed area, the square has the smallest perimeter.

Proof. Let $x, y > 0$ be the side lengths of a rectangle with a fixed area $xy = A$. By the AM-GM inequality,

$$\frac{x+y}{2} \geq \sqrt{xy} = \sqrt{A},$$

so

$$x+y \geq 2\sqrt{A}.$$

The perimeter is $2x+2y$, hence

$$2x+2y \geq 4\sqrt{A}.$$

When the rectangle is a square, $x = y$, the inequality is binding. Therefore, the smallest possible perimeter occurs when $x = y$, that is, when the rectangle is a square. $\square$

A result, once proved, becomes an instrument for proving the next.

3.6. Without Loss of Generality

In Section 2.2, we saw that a proof sometimes has to be divided into cases, each treated on its own. It happens often that two of the cases, once written out, turn out to be the same argument with the variable names switched. When this happens, we may write the argument once and say so. The phrase for saying so is *without loss of generality*. We say that two integers have **opposite parity** if one is even and the other odd.

Proposition 3.6.1

If two integers have opposite parity, then their sum is odd.

Proof. Let m and n be two integers with opposite parity. Without loss of generality, assume that m is even and n is odd. Then, $m = 2a$ for some integer a and $n = 2b + 1$ for some integer b . Hence,

$$m+n = 2a+2b+1 = 2(a+b)+1,$$

and since $a + b$ is an integer, $m + n$ is odd by definition of an odd integer. $\square$

It is easy to see that assuming m odd and n even would change nothing but the names. We have proved both cases by proving one. The phrase is not a way of skipping work; it is a way of not repeating it.

3.7. Proving a Stronger Statement

It sometimes happens that a claim is hard to prove and that a stronger claim is easy. This is less strange than it sounds. When we prove something by mathematical induction, for example, a stronger claim gives us a stronger assumption to carry into the inductive step, and the extra strength may be exactly what we need to carry the argument through.

Suppose, for example, that we want to show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2,$$

for every positive integer n . The base case boils down to $1 \leq 2$, which is certainly true. The inductive hypothesis will be the assumption that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} \leq 2.$$

But now things turn difficult, for in the induction step, we get

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 + \frac{1}{(k+1)^2},$$

which is no good. So we make a stronger claim:

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2 - \frac{1}{n},$$

for all positive integers n . The claim is stronger because $2 - 1/n$ is smaller than 2; so if the sum stays below $2 - 1/n$, then it stays below 2, and the original inequality follows.

Proposition 3.7.1

For all positive integers n ,

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2.$$

Proof. We prove by induction on n the stronger statement

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2 - \frac{1}{n}.$$

For the base case we have $1 \leq 2 - 1 = 1$, which is true. So assume that inequality holds for some k . Then,

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq \left(2 - \frac{1}{k}\right) + \frac{1}{(k+1)^2},$$

and it remains only to check that the right side does not exceed $2 - 1/(k+1)$. The right side is the same as

$$2 - \frac{(k+1)^2}{k(k+1)^2} + \frac{k}{k(k+1)^2} = 2 - \frac{k^2 + 2k + 1 - k}{k(k+1)^2} = 2 - \frac{k^2 + k + 1}{k(k+1)^2}.$$

Because $k^2 + k + 1 > k^2 + k$, the fraction we subtract is larger, and subtracting a larger quantity leaves a smaller result. Hence,

$$< 2 - \frac{k^2 + k}{k(k+1)^2} = 2 - \frac{k(k+1)}{k(k+1)^2} = 2 - \frac{1}{k+1},$$

which is what we set out to show. $\square$

3.8. The Smallest Counterexample

We end with a strategy we have already used, though not by name. When we proved the principle of mathematical induction, we leaned on the plain fact that in any collection of positive integers with at least one member, there is a smallest one. The negation of every positive integer has a property is that some positive integer does not have that property. So suppose for a contradiction, that some does not. The offenders form a collection, and by the fact just noted there is a smallest offender – that is, a smallest number lacking the property. If we can show that such a smallest offender is impossible, then there cannot be any offenders at all, and the property must hold for all.

We illustrate this with a theorem that every student will meet in due course, and which we ourselves took for granted at the beginning of Chapter 2: every positive integer greater than 1 can be written as a product of primes.

Theorem 3.8.1

Every integer $n \geq 2$ can be written as a product of primes.

Proof. Suppose, for a contradiction, that this is false. Then, there is at least one integer $n \geq 2$ that cannot be written as a product of primes. Among all such integers, let m be the smallest. Now m cannot itself be prime, for a prime is already its own product of primes. So m must be composite. Hence, $m = ab$ for some integers a, b with $2 \leq a, b < m$. But m was the smallest integer that cannot be written as a product of primes; so a and b must each have one. Placing the two factorizations together gives a factorization of $m = ab$ into primes, the very thing we supposed m lacked. This is a contradiction. Hence, there is no smallest offender, and so no offender at all: every integer greater than 1 is a product of primes. $\square$

That such a factorization is moreover unique is the harder half of the theorem, and a result the readers now are well equipped to venture after and discover for themselves. The road goes ever on; may yours be a good one.

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